Dynamic Response 3

Professor: Juan David Bastidas Rodríguez

Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

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1. System Modeling Diagrams
2. Effect of Pole Locations
3. Time-Domain Specifications

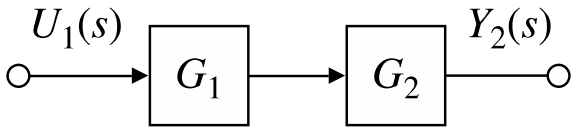
1. System Modeling Diagrams

1. The transfer function of each component is placed in a box, and the input–output relationships between components are indicated by lines and arrows
2. We can solve the equations by graphical simplification
3. Think of each block as representing an electronic amplifier with the transfer function printed inside

1. System Modeling Diagrams

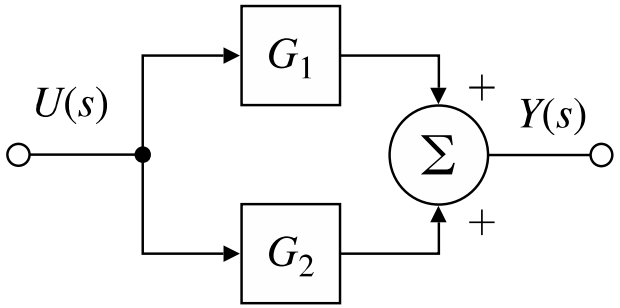
Three examples of elementary block diagrams:

(a) series; (b) parallel; (c) feedback



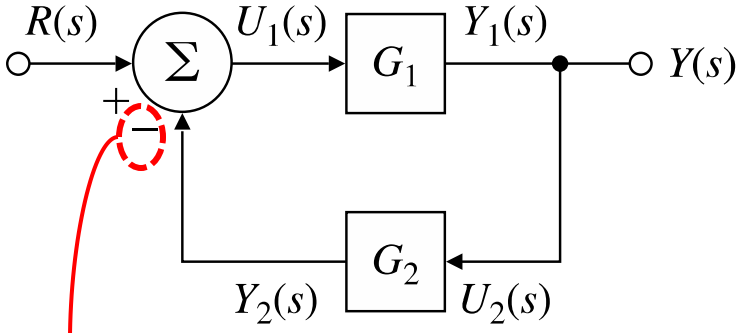
$$\frac{Y_2(s)}{U_1(s)} = G_2G_1$$

(a)



$$\frac{Y(s)}{U(s)} = G_2 + G_1$$

(b)



$$\frac{Y(s)}{R(s)} = \frac{G_1}{1 + G_2G_1}$$

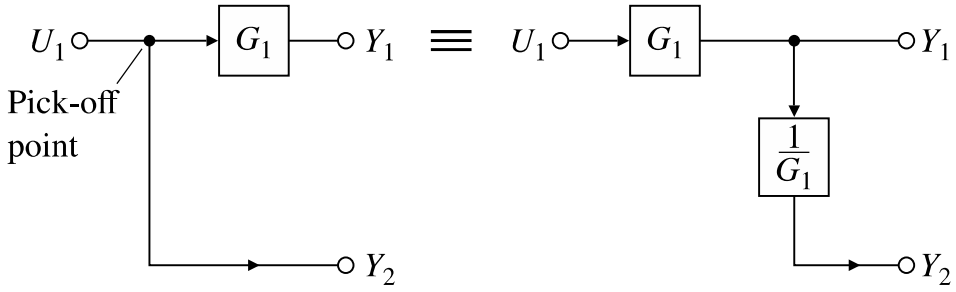
(c)

The gain of a single-loop negative feedback system is given by the forward gain divided by the sum of 1 plus the loop gain.

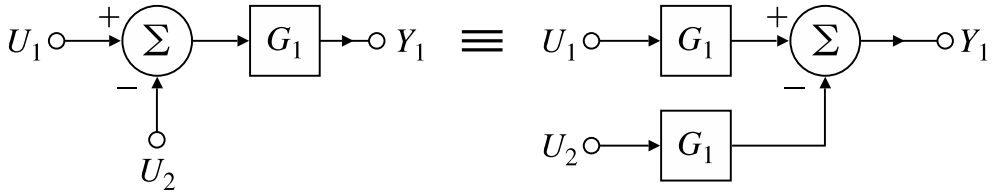
1. System Modeling Diagrams

Examples of block-diagram algebra:

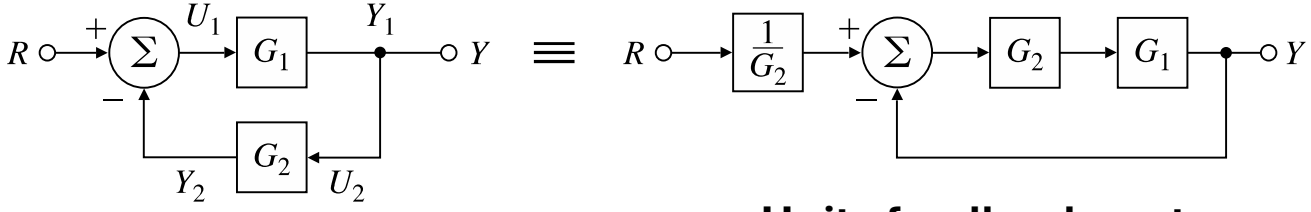
- (a) moving a pickoff point
- (b) moving a summer
- (c) conversion to unity feedback



(a)



(b)



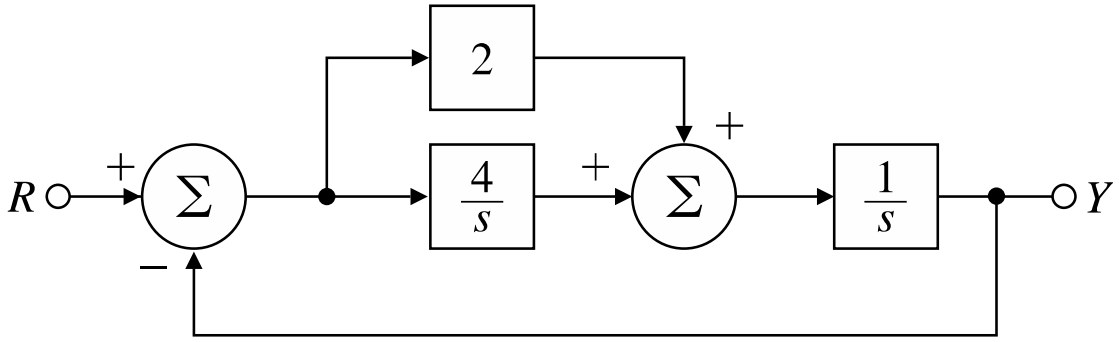
(c)

Unity feedback system

1. System Modeling Diagrams

Example 1: Transfer Function from a Simple Block Diagram

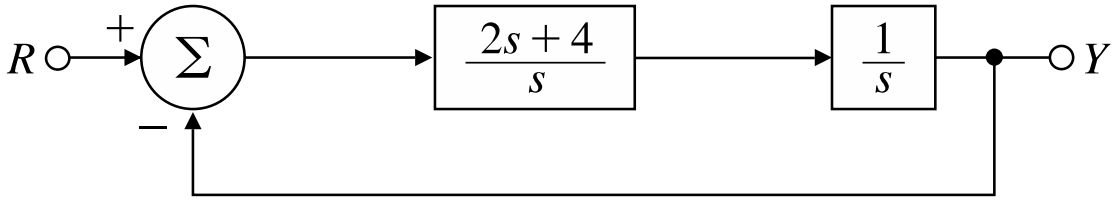
Find the transfer function of the system



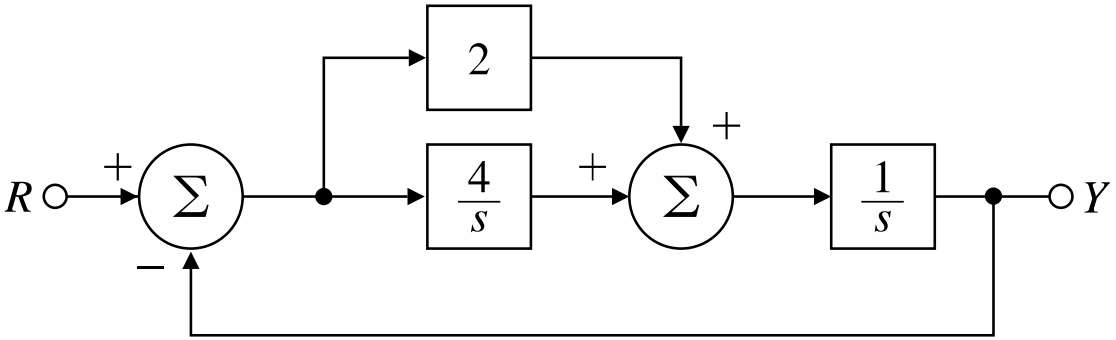
1. System Modeling Diagrams

Example 1: Transfer Function from a Simple Block Diagram

Find the transfer function of the system



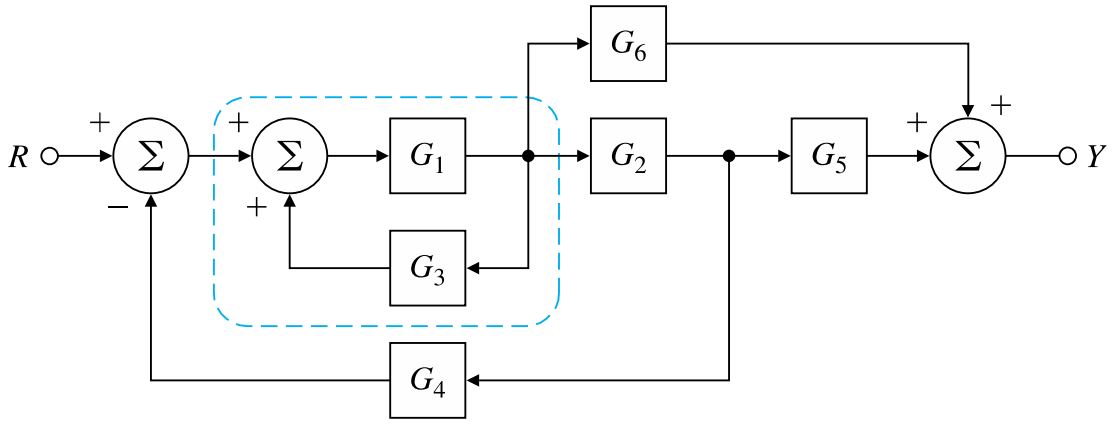
$$T(s) = \frac{Y(s)}{R(s)} = \frac{\frac{2s+4}{s^2}}{1 + \frac{2s+4}{s^2}} = \frac{2s+4}{s^2+2s+4}$$



1. System Modeling Diagrams

Example 2: Transfer Function from the Block Diagram

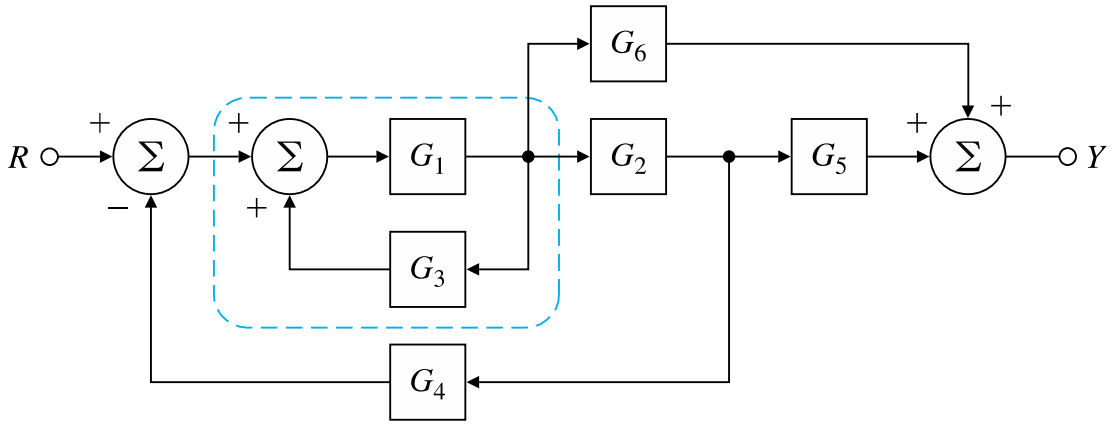
Find the transfer function of the system

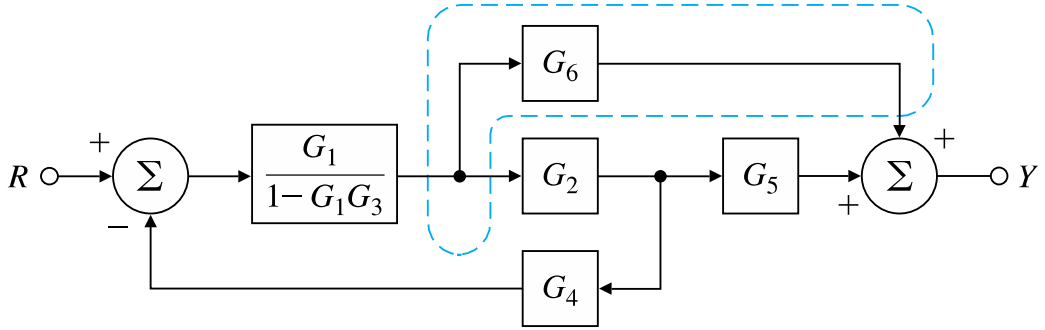


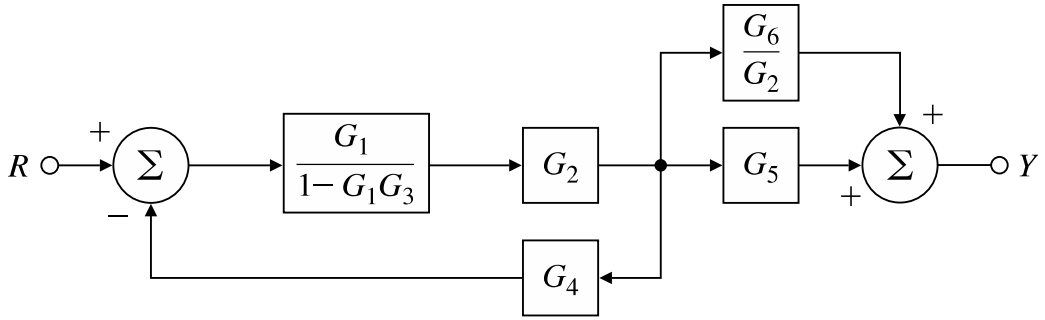
1. System Modeling Diagrams

Example 2: Transfer Function from the Block Diagram

Find the transfer function of the system







$$T(s) = \frac{Y(s)}{R(s)} = \frac{\frac{G_1 G_2}{1-G_1 G_3}}{1 + \frac{G_1 G_2 G_4}{1-G_1 G_3}} \left(G_5 + \frac{G_6}{G_2} \right)$$

$$= \frac{G_1 G_2 G_5 + G_1 G_6}{1 - G_1 G_3 + G_1 G_2 G_4}.$$

1. System Modeling Diagrams

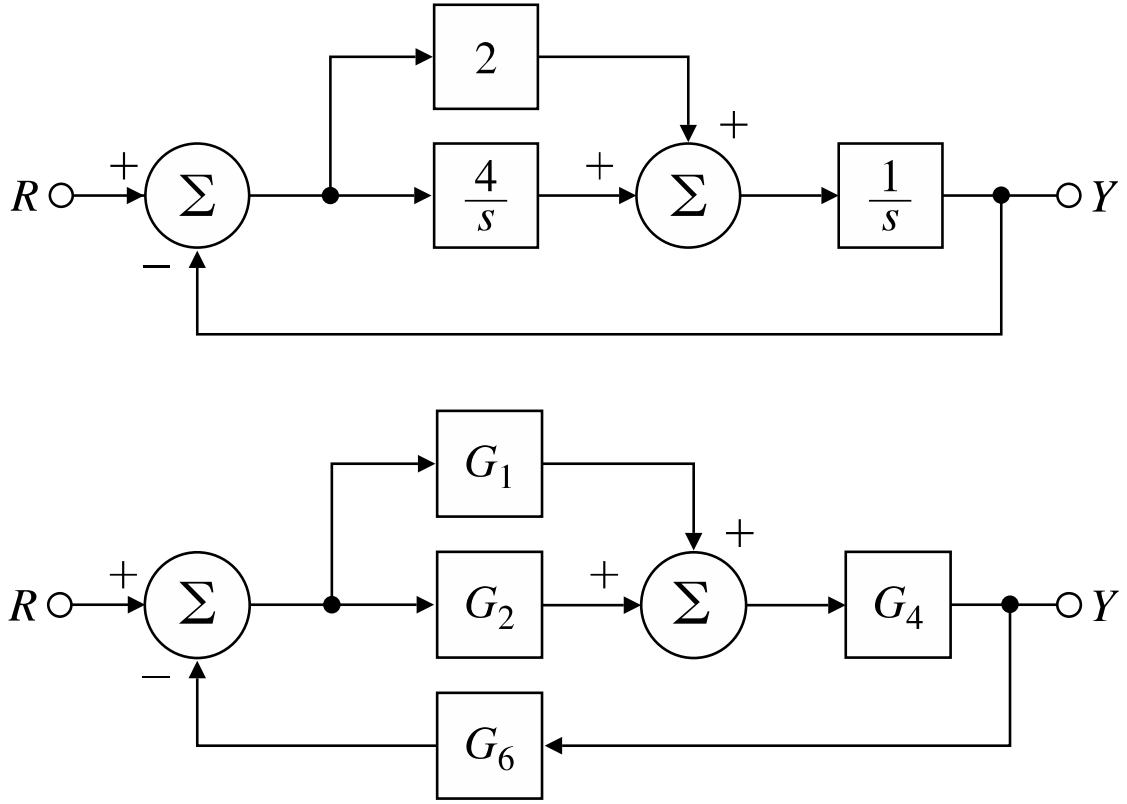
Block-Diagram Reduction Using Matlab

- The three commands:
  - Series
  - Parallel
  - Feedback

1. System Modeling Diagrams

Block-Diagram Reduction Using Matlab

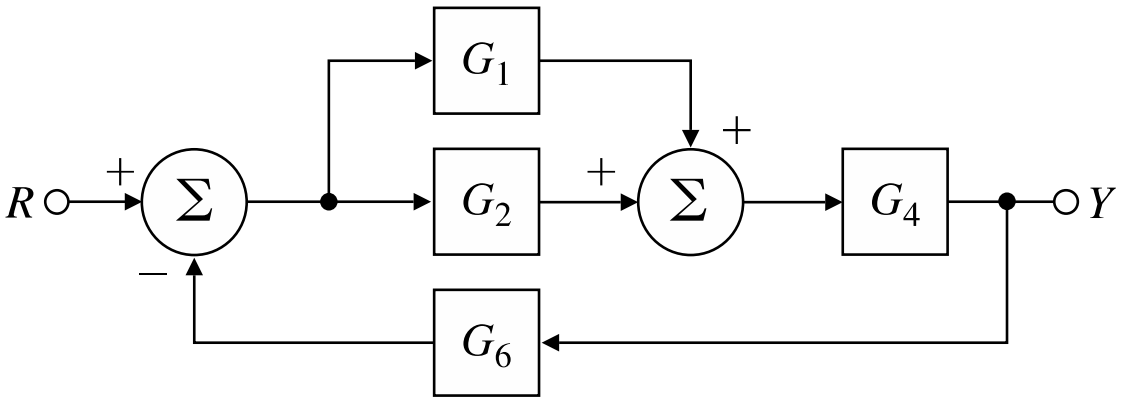
Example 3: repeat the reduction of the system shown in Example 1



1. System Modeling Diagrams

Block-Diagram Reduction Using Matlab

Example 3: repeat the reduction of the system shown in Example 1




```
clear; clc;
s=tf('s'); % specify a transfer function using a rational function in the
% Laplace variable s
sysG1=2; % define subsystem G1
sysG2=4/s; % define subsystem G2
sysG3=parallel(sysG1,sysG2); % parallel combination of G1 and G2
sysG4=1/s; % define subsystem G4
sysG5=series(sysG3,sysG4); % series combination of G3 and G4
sysG6=1;
sysCL=feedback(sysG5,sysG6,-1); % feedback combination of G5 and G6

sysCL =
    2 s + 4
  ------------
  s^2 + 2 s + 4
Continuous-time transfer function.
```

2. Effect of Pole Locations

- Once the transfer function has been determined => we can start to analyze the response of the system **it represents**
- The **poles** and **zeros** completely describe H(s) **except** for a constant multiplier
- **Impulse response** => we call it the **natural response**
- The **poles** identify the **classes of signals** in the **impulse response** (partial-fraction)

2. Effect of Pole Locations

Real pole:

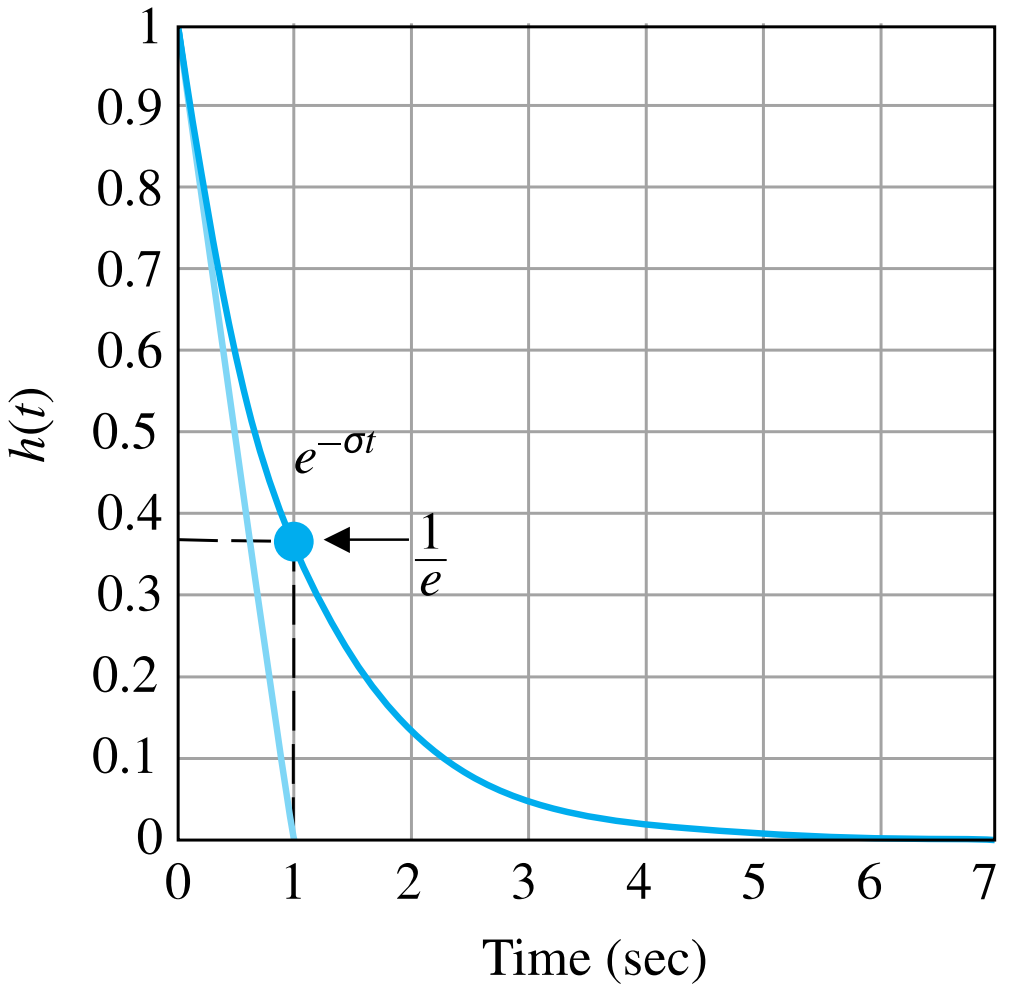
The straight line tangent to the curve at t = 0 and terminates at t = τ can be used to:

- Schetch the time response
- Checking computer results

$$H(s) = \frac{1}{s+\sigma}$$

$$h(t) = e^{-\sigma t}1(t)$$

$$\tau = 1/\sigma$$

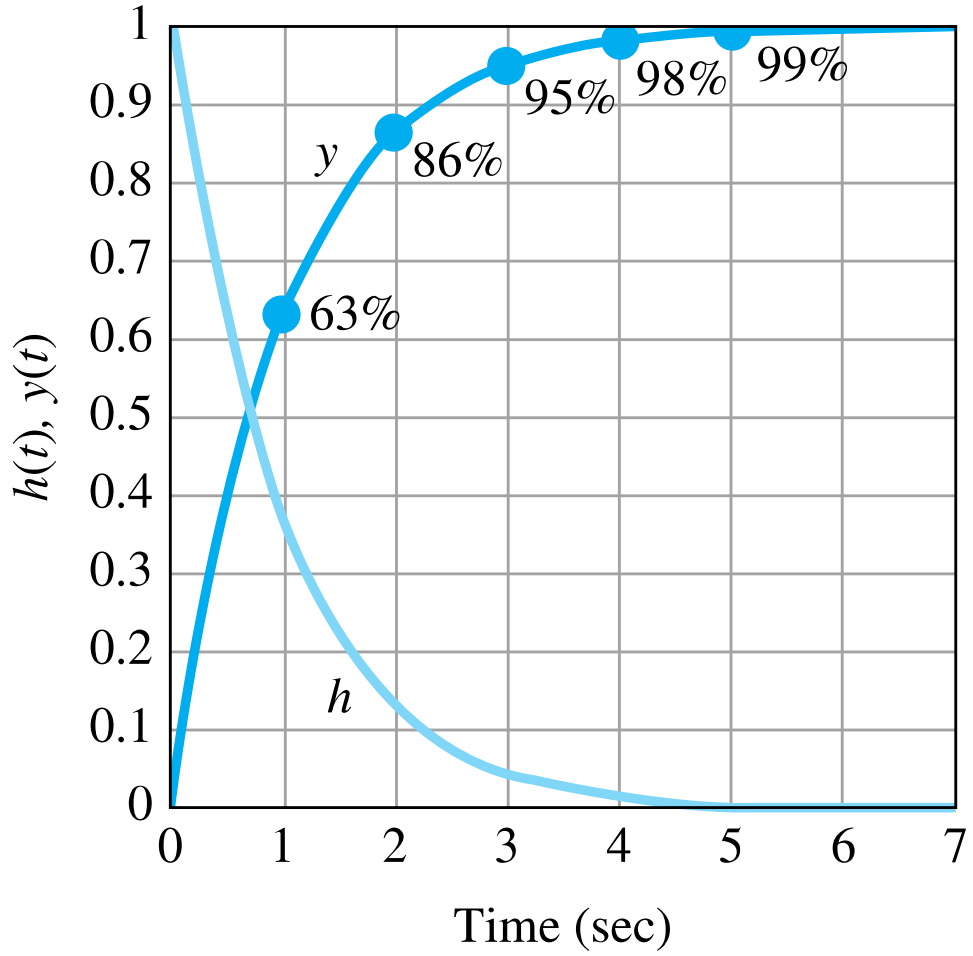


2. Effect of Pole Locations

Real pole:

Impulse and step responses for a first-order system computed using Matlab

- At τ seconds the system reaches 63% of its steady state value
- At 5τ seconds the system is at steady-state



2. Effect of Pole Locations

Response versus Pole Locations, **Real Roots**

Example:

Compare the time response with the pole locations for the system with a transfer function between input and output given by the TF

$$H(s) = \frac{2s + 1}{s^2 + 3s + 2}$$

2. Effect of Pole Locations

Response versus Pole Locations, **Real Roots**

Example:

Compare the time response with the pole locations for the system with a transfer function between input and output given by the TF

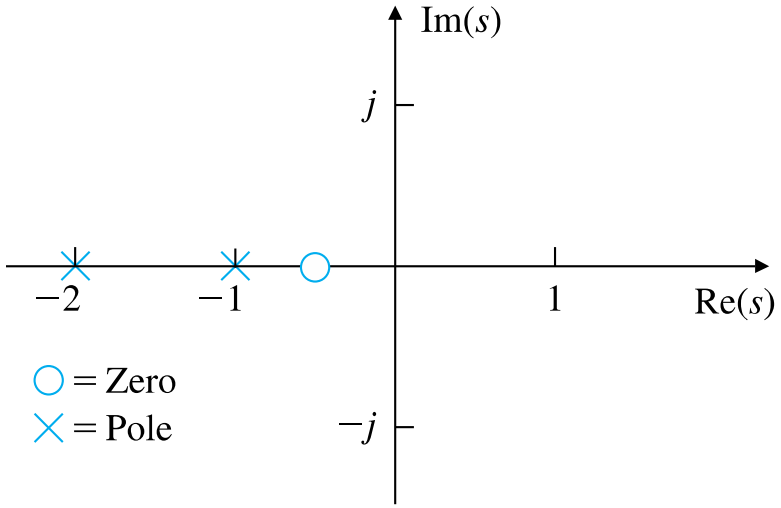
$$H(s) = \frac{2s+1}{s^2+3s+2}$$

Numerator

$$b(s) = 2\left(s+\frac{1}{2}\right)$$

$$a(s) = s^2+3s+2 = (s+1)(s+2)$$

Denominator



Partial fraction

$$H(s) = -\frac{1}{s+1} + \frac{3}{s+2}$$

$$h(t) = \begin{cases} -e^{-t}+3e^{-2t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

2. Effect of Pole Locations

Response versus Pole Locations, **Real Roots**

Example:

Compare the time response with the pole locations for the system with a transfer function between input and output given by the TF

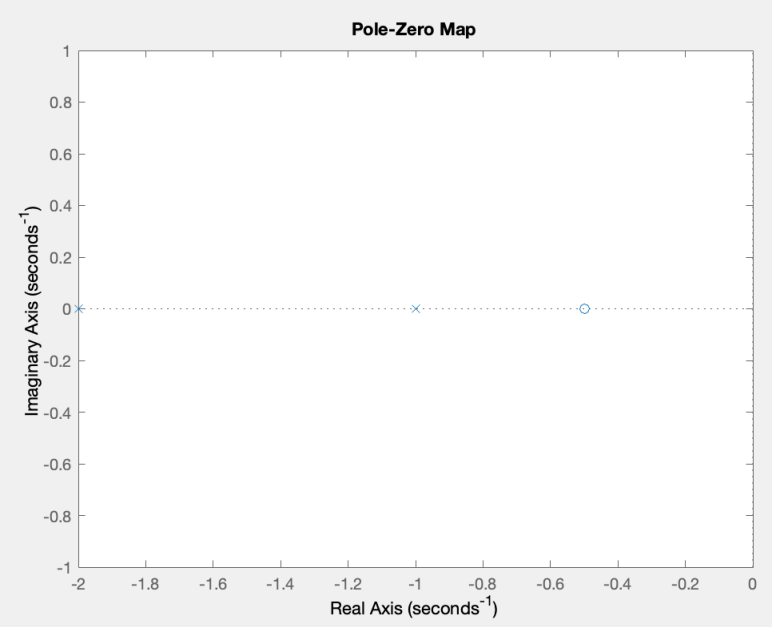
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Numerator

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$$a(s) = s^2+3s+2 = (s+1)(s+2)$$

Denominator




```
num=[2 1];
den=[1 3 2];
pzmap(num,den);
```

2. Effect of Pole Locations

Response versus Pole Locations, **Real Roots**

Example:

Compare the time response with the pole locations for the system with a transfer function between input and output given by the TF

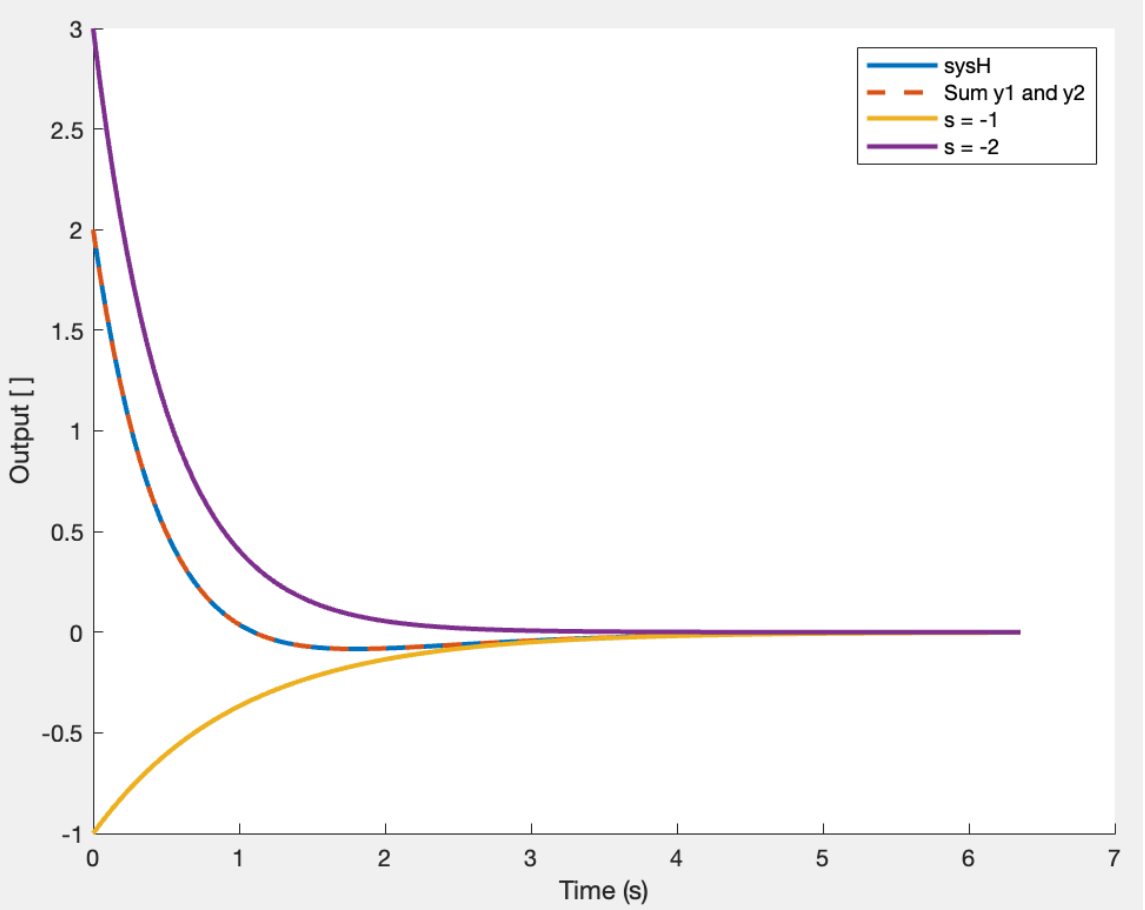
```
s=tf('s'); % define Laplace variable
sysH=(2*s+1)/(s^2+3*s+2); % define system from its numerator and
% denominator
[y,t] = impulse(sysH); % compute impulse response
sys1 = -1/(s+1);
sys2 = 3/(s+2);
[y1,t1] = impulse(sys1,t);
[y2,t2] = impulse(sys2,t);
figure(2); hold on;
plot(t,y,'DisplayName','sysH','LineWidth',2);
plot(t1,y1+y2,'--','DisplayName','Sum y1 and y2','LineWidth',2);
plot(t1,y1,'DisplayName','s = -1','LineWidth',2);
plot(t2,y2,'DisplayName','s = -2','LineWidth',2);
xlabel('Time (s)'); ylabel('Output [ ]');
legend; hold off;
```

2. Effect of Pole Locations

Response versus Pole Locations, **Real Roots**

Example:

Compare the time response with the pole locations for the system with a transfer function between input and output given by the TF



2. Effect of Pole Locations

Complex poles:

- A pole has a negative real part if σ is positive
- Denominator associated
- From a second-order differential equation we arrive to =>
- ζ is the damping ratio
- ωn is the undamped natural frequency

$$s = -\sigma \pm j\omega_d$$

$$a(s) = (s + \sigma - j\omega_d)(s + \sigma + j\omega_d) = (s + \sigma)^2 + \omega_d^2$$

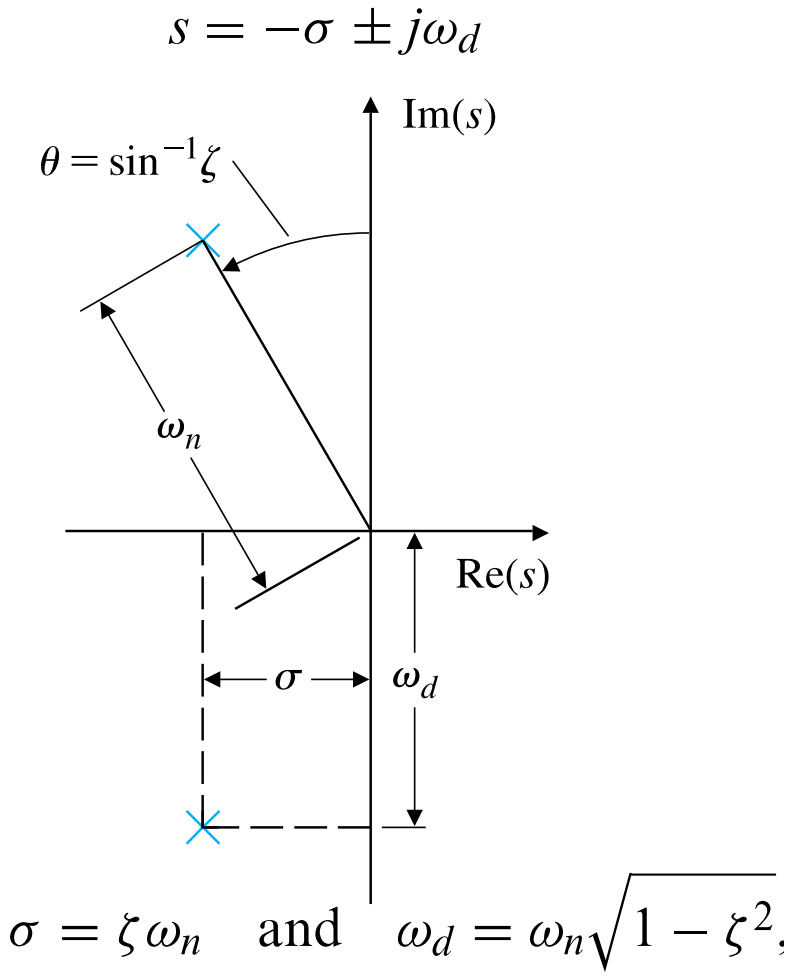
$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\sigma = \zeta\omega_n \quad \text{and} \quad \omega_d = \omega_n\sqrt{1 - \zeta^2}$$

2. Effect of Pole Locations

Complex poles:

- A pole has a negative real part if σ is positive
- Denominator associated
- From a second-order differential equation we arrive to =>
- ζ is the damping ratio
- ωn is the undamped natural frequency



2. Effect of Pole Locations

Complex poles:

$$s = -\sigma \pm j\omega_d$$

- Time response of H(s)

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- Re-writing H(s)

$$H(s) = \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

- Using the table for inverse Laplace transform

$$\frac{b}{(s+a)^2 + b^2} \qquad e^{-at}\sin bt$$

$$h(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\sigma t}(\sin \omega_d t)1(t)$$

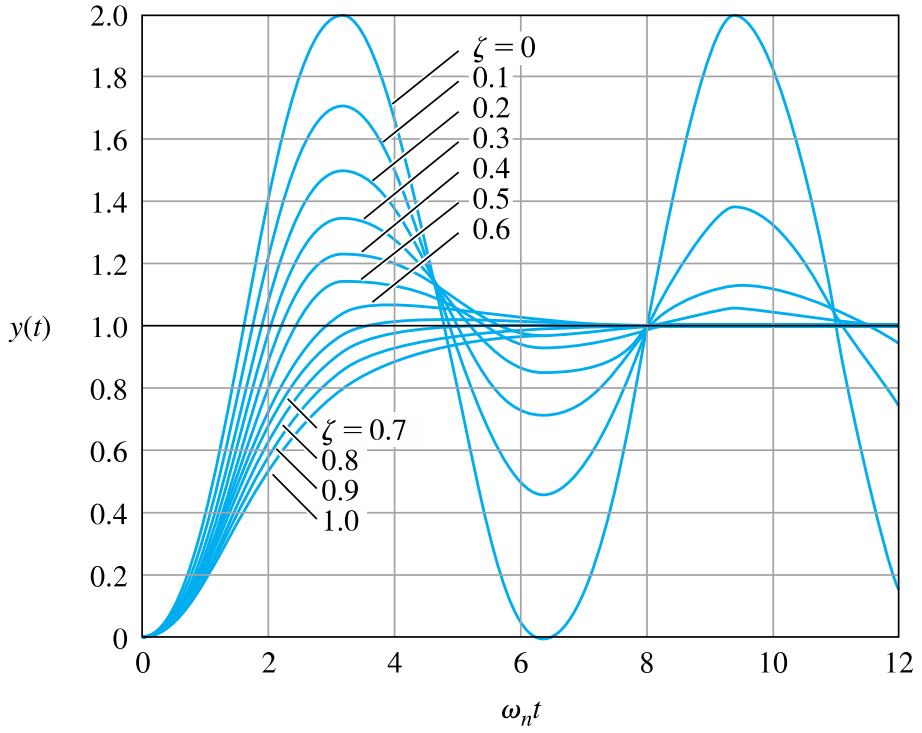
2. Effect of Pole Locations

Complex poles:

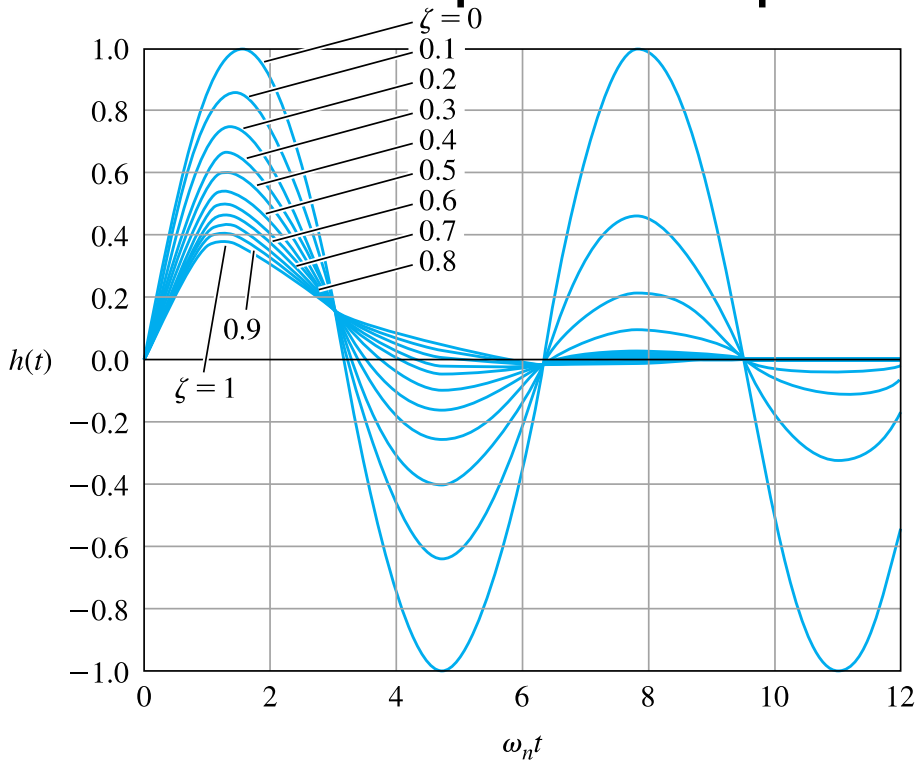
$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$h(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\sigma t} (\sin \omega_d t) 1(t)$$

Step response



Impulse response

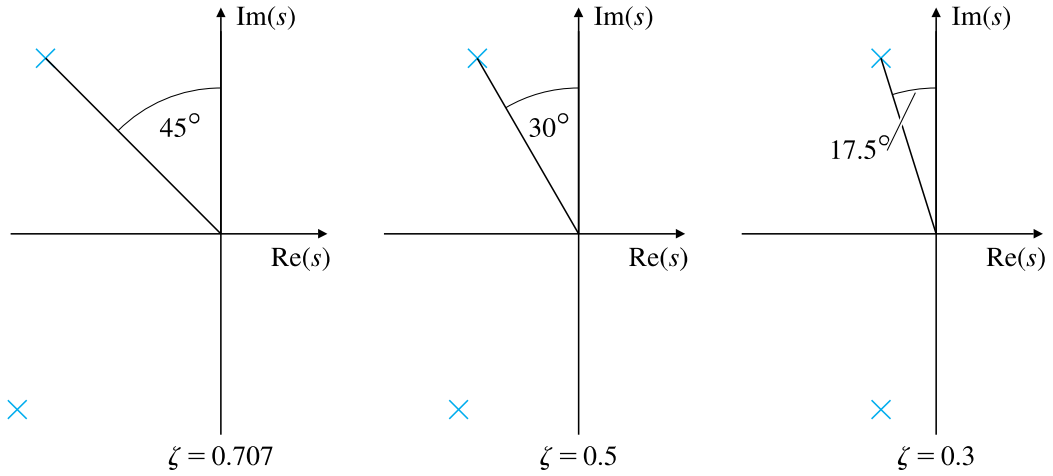


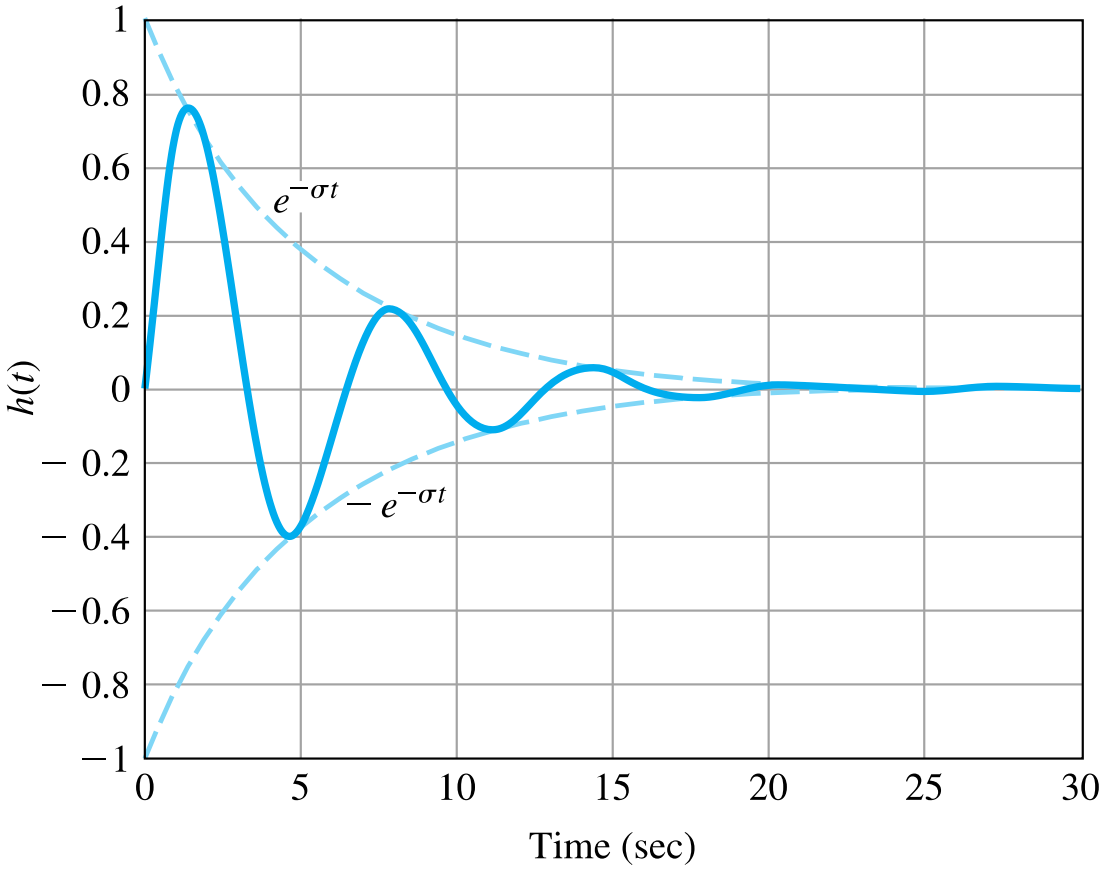
2. Effect of Pole Locations

Complex poles:

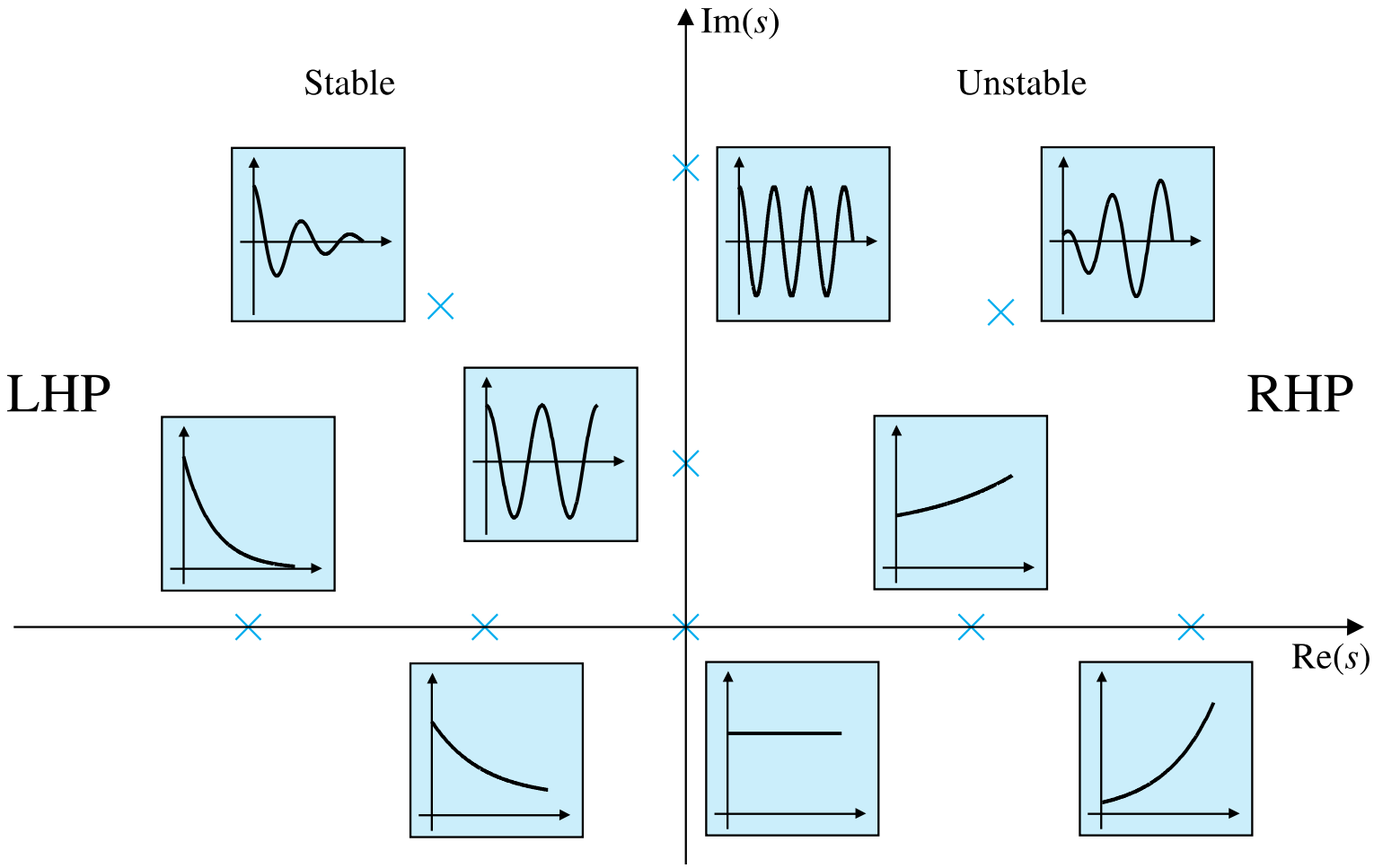
$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$h(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\sigma t} (\sin \omega_d t) 1(t)$$





2. Effect of Pole Locations



2. Effect of Pole Locations

$$H(s) = \frac{2s+1}{s^2+2s+5}$$

Complex poles

Example : Oscillatory Time Response

Discuss the correlation between the poles of H(s) and find the impulse response

Inverse Laplace transform from tables

$$\frac{s+a}{(s+a)^2+b^2} \qquad e^{-at}\cos bt$$

$$\frac{b}{(s+a)^2+b^2} \qquad e^{-at}\sin bt$$

2. Effect of Pole Locations

Complex poles

Example : Oscillatory Time Response

Discuss the correlation between the poles of H(s) and find the impulse response

$$H(s) = \frac{2s+1}{s^2+2s+5}$$

$$\omega_n^2 = 5 \Rightarrow \omega_n = \sqrt{5} = 2.24 \text{ rad/sec}$$

$$2\zeta\omega_n = 2 \Rightarrow \zeta = \frac{1}{\sqrt{5}} = 0.447$$

$$H(s) = \frac{2s+1}{s^2+2s+5} = \frac{2s+1}{(s+1)^2+2^2}$$

$$H(s) = \frac{2s+1}{(s+1)^2+2^2} = 2\frac{s+1}{(s+1)^2+2^2} - \frac{1}{2}\frac{2}{(s+1)^2+2^2}$$

Inverse Laplace transform from tables

$\frac{s+a}{(s+a)^2+b^2}$	$e^{-at}\cos bt$
$\frac{b}{(s+a)^2+b^2}$	$e^{-at}\sin bt$

2. Effect of Pole Locations

Complex poles

Example : Oscillatory Time Response

Discuss the correlation between the poles of H(s) and find the impulse response

$$H(s) = \frac{2s+1}{s^2+2s+5}$$

$$h(t) = \left(2e^{-t}\cos 2t - \frac{1}{2}e^{-t}\sin 2t\right)1(t)$$

```
s=tf('s'); % define Laplace variable
sysH=(2*s+1)/(s^2+2*s+5); % define system by its numerator and denominator
t=0:0.1:6; % form time vector
y=impulse(sysH,t); % compute impulse response
plot(t,y); % plot impulse response
```

2. Effect of Pole Locations

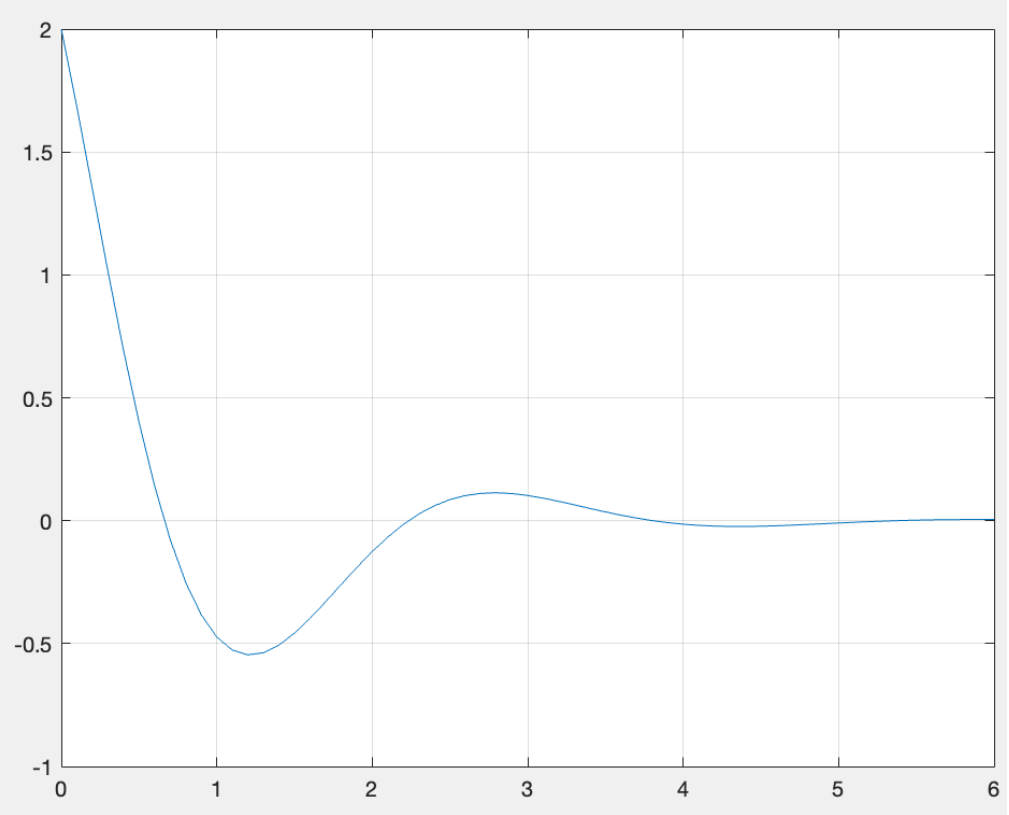
Complex poles

Example : Oscillatory Time Response

Discuss the correlation between the poles of H(s) and find the impulse response

$$H(s) = \frac{2s+1}{s^2+2s+5}$$

$$h(t) = \left(2e^{-t}\cos 2t - \frac{1}{2}e^{-t}\sin 2t\right)1(t)$$



3. Time-Domain Specifications

For a step response:

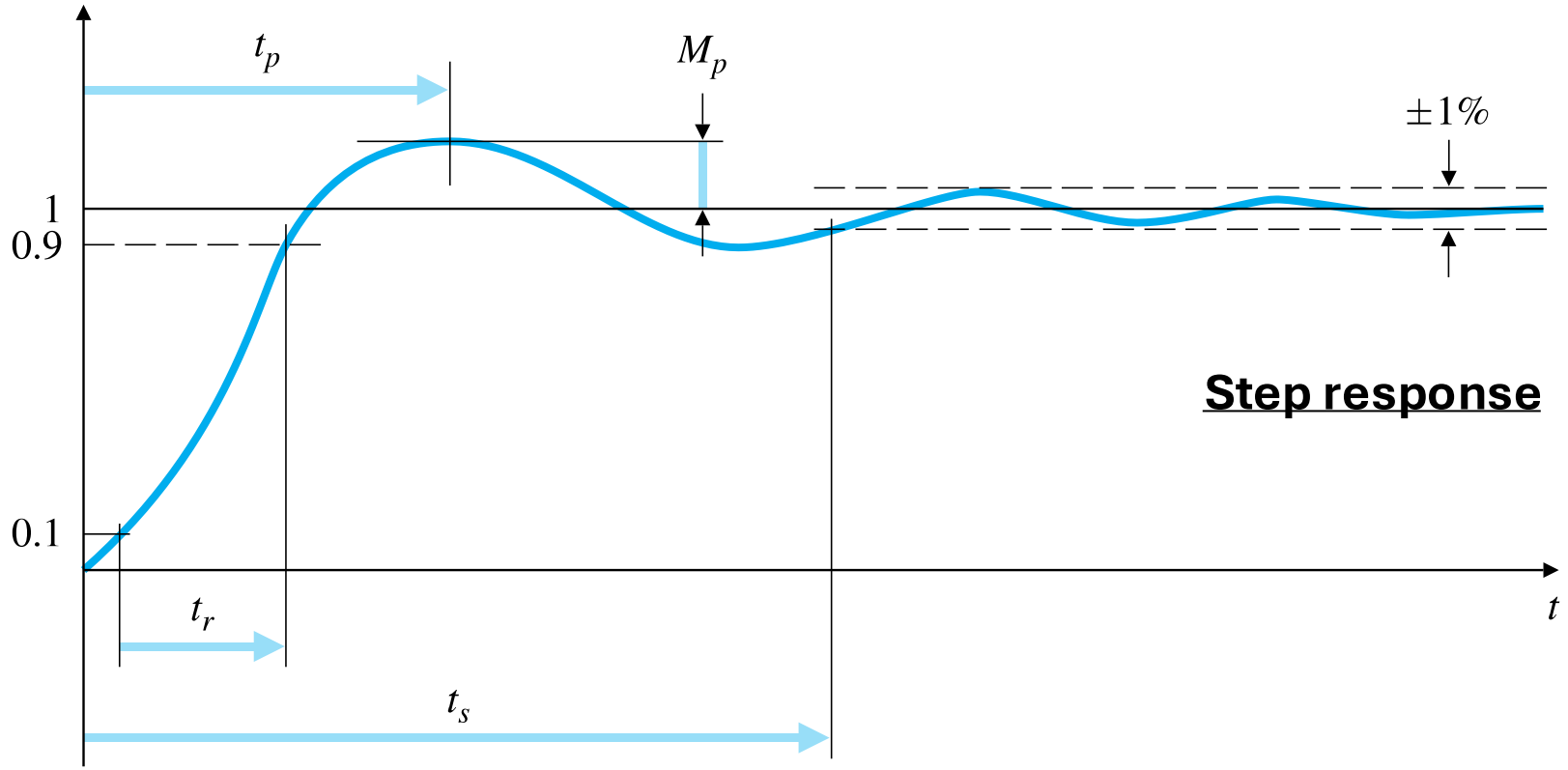
1. The **rise time tr** is the time it takes the system to reach the vicinity of its new set point.
2. The **settling time ts** is the time it takes the system transients to decay.
3. The **overshoot Mp** is the maximum amount the system overshoots its final value divided by its final value (and is often expressed as a percentage).
4. The **peak time tp** is the time it takes the system to reach the maximum overshoot point.

3. Time-Domain Specifications

Rise time approximation: for a 2nd order system with $\zeta = 0.5$ and no zeros

$$t_r \cong \frac{1.8}{\omega_n}$$

Only a rough approximation



Step response

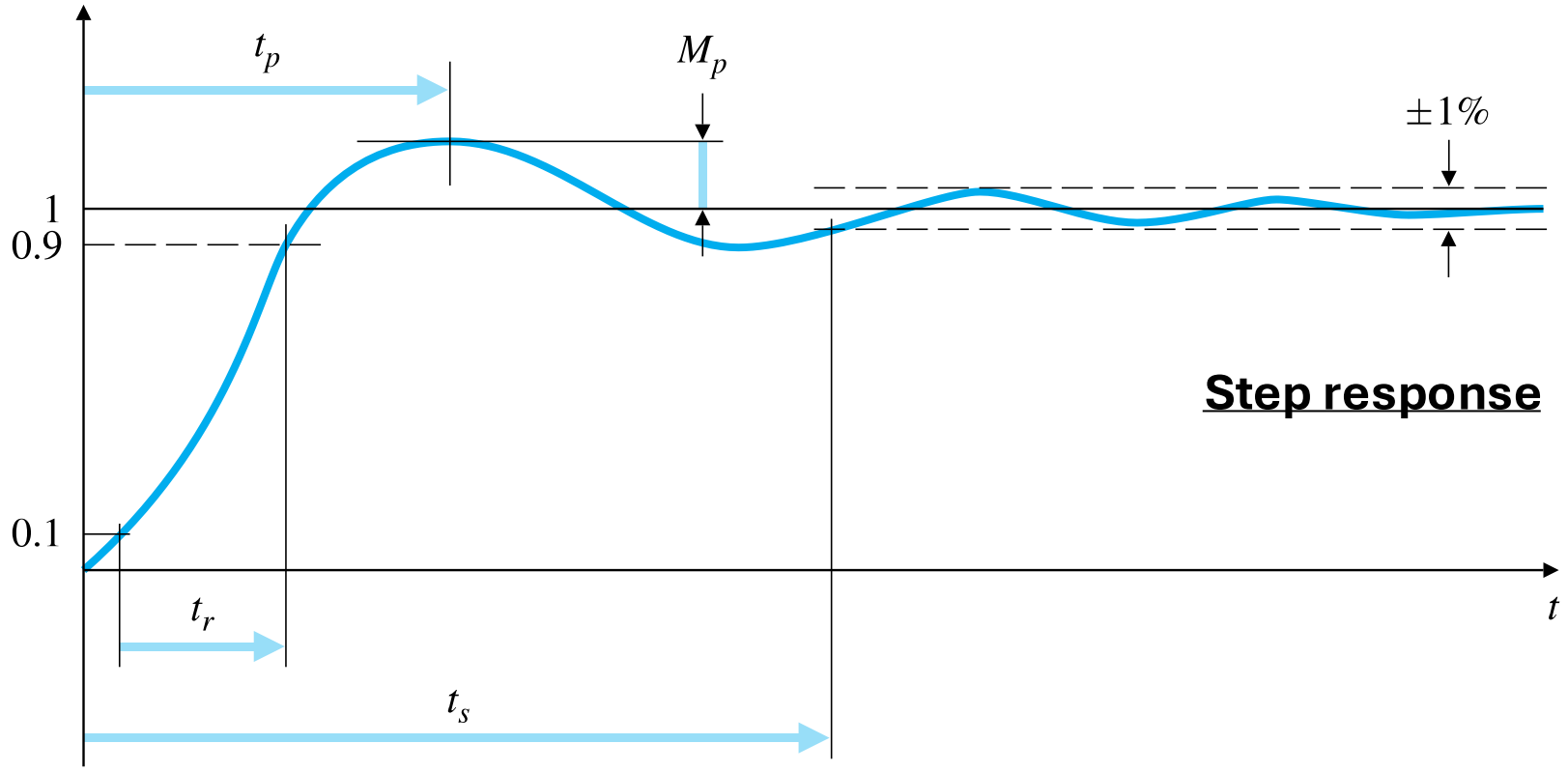
3. Time-Domain Specifications

Overshoot and peak time: for a 2nd order system and no zeros

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \quad 0 \leq \zeta < 1$$

$$t_p = \frac{\pi}{\omega_d}$$

Obtained analytically



Step response

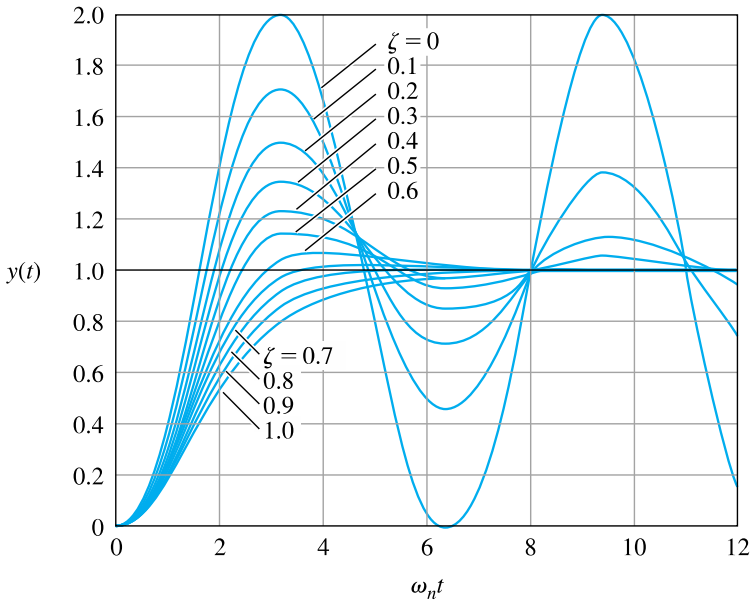
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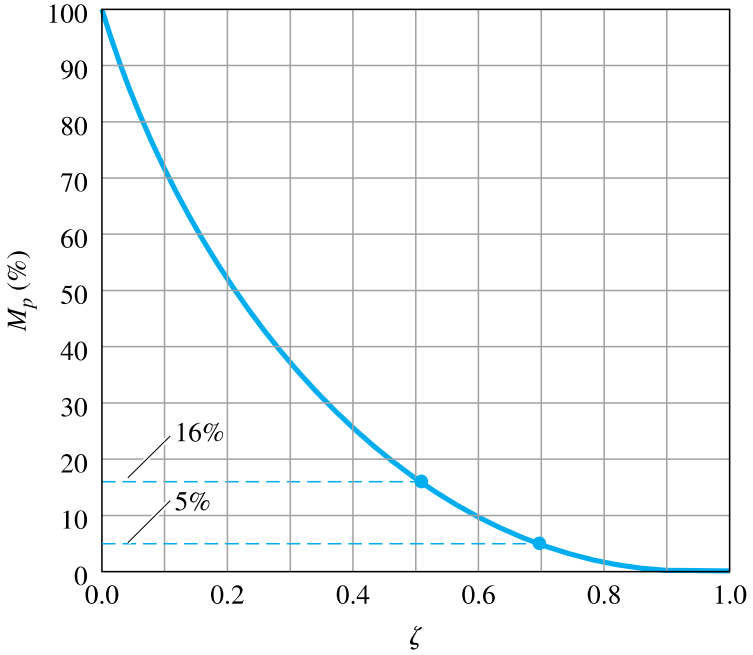
Overshoot and peak time: for a 2nd order system and no zeros

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \quad 0 \leq \zeta < 1$$

$$t_p = \frac{\pi}{\omega_d}$$

Mp for different values of ζ





3. Time-Domain Specifications

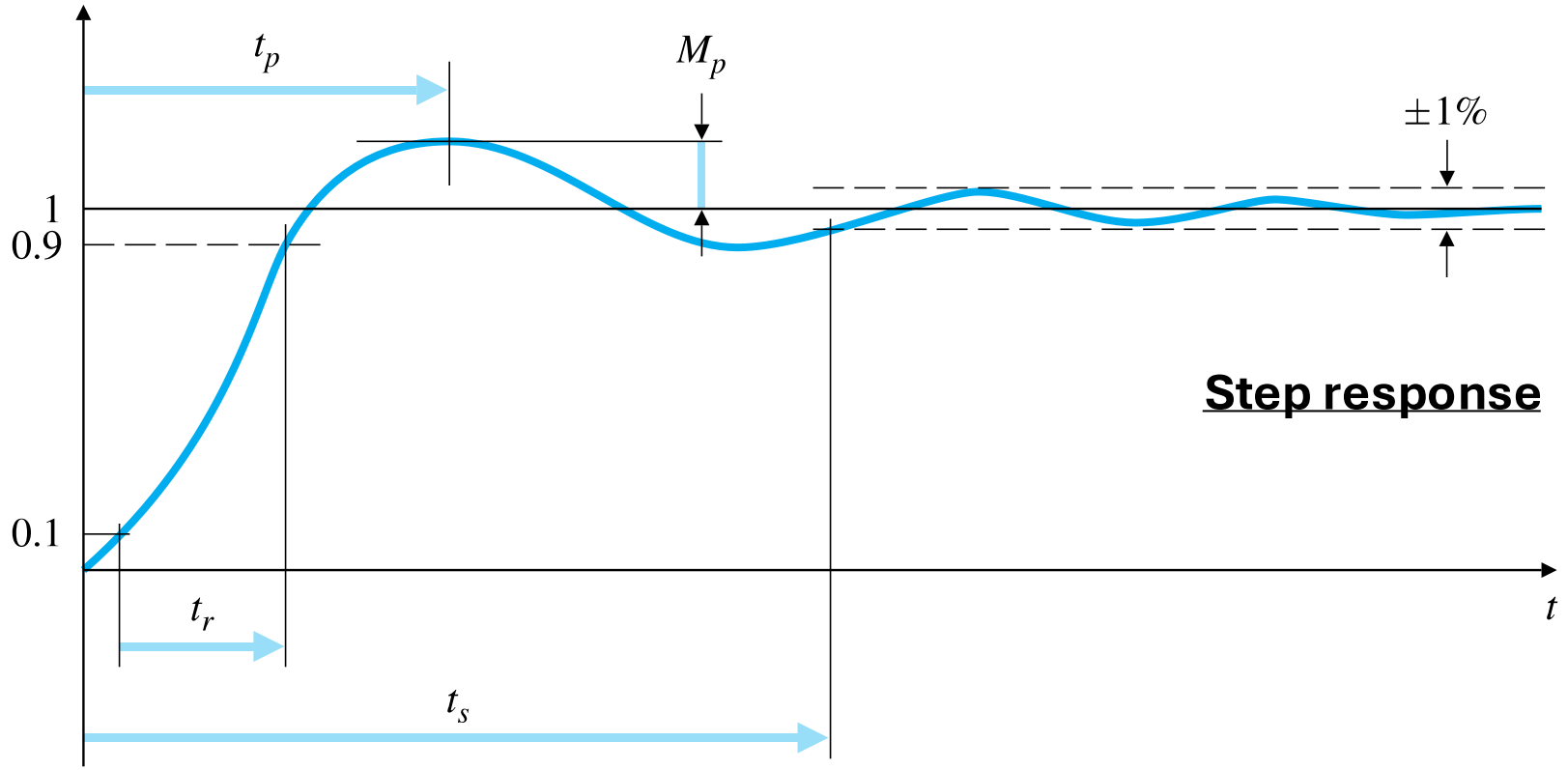
Settling time: for a 2nd order system and no zeros

1% criterion

$$e^{-\zeta\omega_n t_s} = 0.01$$

$$\zeta\omega_n t_s = 4.6,$$

$$t_s = \frac{4.6}{\zeta\omega_n} = \frac{4.6}{\sigma}$$



Step response

What is the criterion for 2% and 5%?

3. Time-Domain Specifications

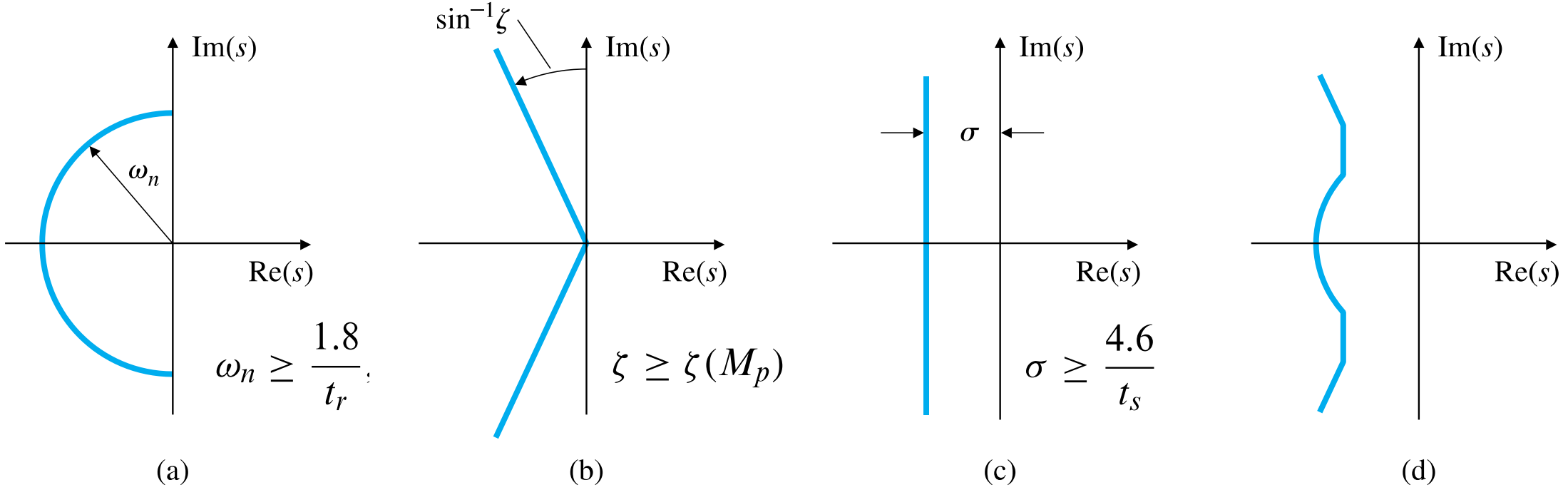


Figure 3.25

Graphs of regions in the *s*-plane delineated by certain transient requirements: (a) rise time; (b) overshoot; (c) settling time; (d) composite of all three requirements

3. Time-Domain Specifications

Simpler for a 1st order system:

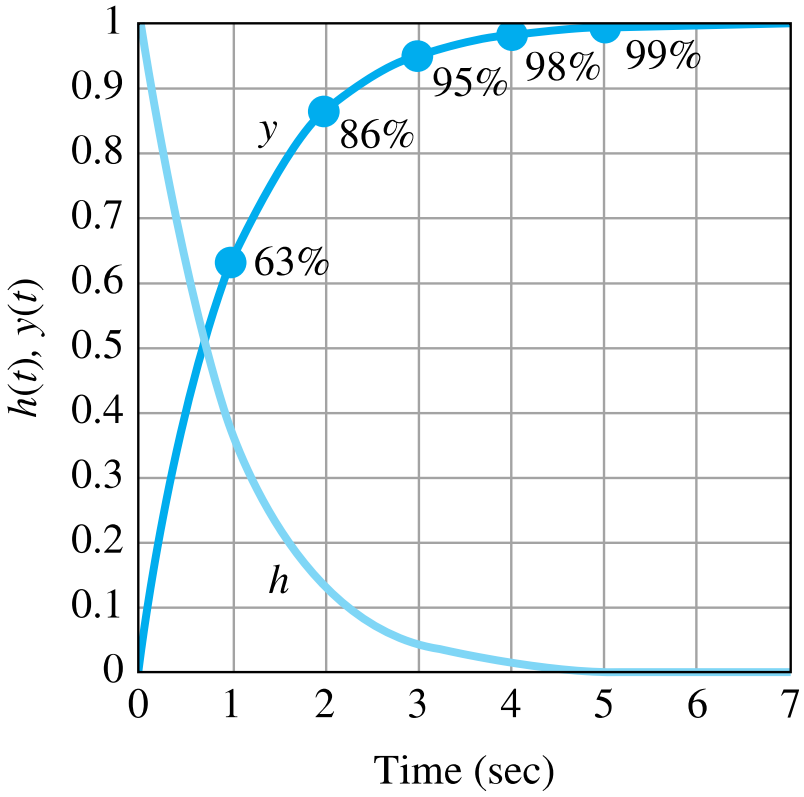
$$H(s) = \frac{\sigma}{s+\sigma}$$

Step response $$Y(s) = \frac{\sigma}{s(s+\sigma)}$$

Step response $$y(t) = (1 - e^{-\sigma t})1(t)$$

Settling time 1% criterion $$t_s = \frac{4.6}{\sigma}$$

Rise time $$t_r = \frac{\ln 0.9 - \ln 0.1}{\sigma} = \frac{2.2}{\sigma}$$



3. Time-Domain Specifications

Example: Transformation of the Specifications to the s-Plane

Find the allowable regions in the s-plane for the poles of the transfer function of the system if the system response requirement is tr ≤ 0.6 sec, Mp ≤ 10%, and ts ≤ 3 sec

$$\omega_n \geq \frac{1.8}{t_r}, \qquad \zeta \geq \zeta(M_p) \qquad \sigma \geq \frac{4.6}{t_s}$$

3. Time-Domain Specifications

Example: Transformation of the Specifications to the s-Plane

Regions in the s-plane for tr ≤ 0.6 sec, Mp ≤ 10%, and ts ≤ 3 sec

$$\omega_n \geq \frac{1.8}{t_r} = 3.0 \text{ rad/sec}$$

$$\zeta \geq 0.6,$$

$$\sigma \geq \frac{4.6}{3} = 1.5 \text{ sec.}$$

